# YASAMAN HAJNOROUZALI

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## Research Profile

Ph.D. researcher in autonomous driving with hands-on experience in motion planning, model predictive control, learning-based driving, closed-loop simulation, and real-world vehicle deployment. Experienced in diagnosing vehicle behavior using logs and simulation, designing controlled experiments, and translating failure analysis into algorithm and controller improvements.

## Education

**Ph.D., Mechanical & Aerospace Engineering** **Aug 2023 – August 2027 (expected)**
*University of Oklahoma* *Norman, OK*
- Full-time Ph.D. candidate; research emphasis in autonomous driving, robotics, learning-based planning, and optimization.

**M.S., Mechanical & Aerospace Engineering** **Jan 2022 – Jul 2023**
*University of California, Davis* *Davis, CA*
- Coursework: Machine Learning Applications, MIMO Control, Advanced Process Control, Sensors & Actuators.

**M.S., Aerospace Engineering** **2016 – 2020**
*University of Isfahan* *Isfahan, Iran*
- Ranked 2nd by GPA in graduating class; thesis on finite-time control and disturbance observation with HIL validation.

## Technical Skills

**Programming & Embedded Systems:** C/C++, Python, MATLAB/Simulink, Linux, embedded firmware, CAN bus, Git
**Control & Robotics:** model predictive control, nonlinear control, motion planning, HIL simulation, ROS/ROS2, sensors and actuators, GPS, LiDAR, camera integration
**Machine Learning & Deployment:** PyTorch, Transformer architectures, end-to-end learning, imitation learning, ONNX, model evaluation
**Research Interests:** embedded systems and control, model predictive control, robotics, autonomous systems, hardware-in-the-loop validation

## Research Experience

**Graduate Research Assistant – End-to-End Autonomous Driving** **Aug 2023 – Present**
*Mobility Intelligence Laboratory (MiLA), University of Oklahoma* *Norman, OK*
- Developed an end-to-end autonomous-driving pipeline using real-world data collected with a Comma 3X on a Nissan Leaf, spanning GPS-derived supervision, PyTorch Transformer training, ONNX export, and on-vehicle deployment at 5 Hz.
- Replaced self-reinforced curvature labels with geometry-based GPS ground truth, improving prediction accuracy over the original labeling pipeline (Pearson $r = 0.805$ on held-out routes); evaluated performance using MAE, RMSE, correlation, and route-level trajectory analysis.
- Designed closed-loop and open-loop kinematic MPC simulations against recorded GPS routes and conducted planning-horizon sweeps to diagnose optimizer behavior, distinguishing steering-sign, latency, and reference-alignment errors.
- Investigated intersection behavior, turn geometry, and lookahead control, identifying failure modes such as wide turns and delayed steering, and translated findings into model and controller evaluation criteria.
- Improved on-road robustness by validating steering-sign conventions, applying curvature limits and exponential smoothing, and correcting the previous-curvature feedback path to reduce curvature oscillation and improve command stability.
- Implemented a partial-model deployment strategy that preserved non-curvature policy outputs while replacing only the curvature prediction branch, preventing invalid trajectory and interface outputs.
- Authored a first-author SAE 2026 paper on curvature-based end-to-end autonomous-vehicle path planning at intersections.

**Graduate Research Assistant – Nonlinear Control & Hardware-in-the-Loop** **Jan 2018 – Jan 2020**
*University of Isfahan* *Isfahan, Iran*
- Designed a super-twisting attitude controller combined with a nonlinear disturbance observer (NDO) for a flexible spacecraft under external disturbances and actuator saturation, comparing four output-definition strategies via Lyapunov-based finite-time stability analysis.
- Reduced attitude settling time by **∼30%** (624s to 440s) versus the baseline super-twisting controller by incorporating the NDO, while maintaining bounded panel deflection and control effort within actuator limits; published in the *Journal of Vibration and Control* (2023).

- Implemented and validated the composite controller on a hardware-in-the-loop spacecraft attitude simulator with reaction-wheel actuation, comparing simulation and experimental results under uncertain inertia and disturbance conditions; published in *Asian Journal of Control* (2023).

## Teaching Experience

**Graduate Teaching Assistant – Robotics, Controls & Engineering Computing** **Aug 2023 – Present**
*University of Oklahoma* *Norman, OK*
- Led laboratory and recitation sessions in robotics, control systems, mechanical engineering, and scientific computing.
- Mentored students in Python, MATLAB, system modeling, experimental data analysis, debugging, and technical reporting.
- Supported laboratory setup, office hours, grading, and assessment for undergraduate engineering courses.

**Graduate Teaching Assistant – Mechanical & Aerospace Engineering** **Jan 2022 – Jul 2023**
*University of California, Davis* *Davis, CA*
- Supported instruction in dynamics, controls, engineering computation, and laboratory-based coursework.
- Guided students in MATLAB, mathematical modeling, data analysis, and engineering problem-solving.

## Selected Research Projects

**Autonomous Navigation, Planning & State Estimation** **2022 – 2023**
- Built autonomous mobile-robot pipelines using GPS, compass, LiDAR, Extended Kalman filtering, A*, Hybrid A*, RRT*, and linearized MPC; evaluated planning and control tradeoffs in simulation.

**NormalNet: Deep Learning-Based Normal Estimation for 3D Point Clouds** **2022**
- Extended PointNet by replacing its max-pooling symmetric function with a sum-based aggregation and concatenating local and global features as input to a six-layer fully-connected network for point-wise normal estimation.
- Trained and evaluated on the PCPNet dataset using a cosine-similarity loss, reducing test loss from **0.063** (global features only) to **0.043** (combined local and global features with sum aggregation).

**MIMO Pole-Placement Controller Design for Lithium-Ion Battery Charging** **2022**
- Derived a nonlinear equivalent-circuit battery model and linearized it into a state-space regulator problem; designed a state-feedback controller via MIMO pole placement to safely regulate terminal voltage during charging.
- Verified closed-loop stability and controllability/observability, and validated controller robustness against ECM parameter variation (temperature and state-of-charge dependence) in Simulink.

## Peer-Reviewed Publications

- Hajnorouzali, Y., et al. "Curvature-Based End-to-End Autonomous Vehicle Path Planning at Intersections." *SAE International*, 2026. **First author**.
- Hajnorouzali, Y., et al. "Finite-time disturbance observer based-control of flexible spacecraft." *Journal of Vibration and Control*, 2023. **First author**.
- Hajnorouzali, Y., et al. "Finite-time control of spacecraft and hardware in the loop tuning." *Asian Journal of Control*, 2022. **First author**.

## Reviewing & Leadership

- **Peer Reviewer**: Aerospace Science and Technology, European Journal of Control, Journal of Vibration and Control, Advances in Space Research, and Journal of Electric Vehicles.
- **President**, Iranian Student Association (ISA), University of Oklahoma (2024 – 2025).
- **Secretary**, Iranian Student Association (ISA), University of California, Davis (2022 – 2023).

## Honors

OU Graduate Fellowship | IAWF Scholarship | UC Davis Graduate Fellowship | Ranked 680/∼40,000 in Iran M.S. entrance exam